MATE70020
Modelling Materials with Density Functional Theory

2026 Fall

Lecture 2

PSet 1 due Oct 27 (in one week)

Calculations of the H_2 molecule

https://canvas.imperial.ac.uk/courses/10590

Recap

Schrödinger's equation

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(\vec{r})\psi = \varepsilon\psi \quad \text{or} \quad \hat{H}\psi = \varepsilon\psi$$

- Electron in free space

$$V(\vec{r}) = 0 \qquad \psi \propto e^{i\vec{k}\cdot\vec{r}}$$

- Electron in a hydrogen atom

$$\psi \propto e^{-r/a_0} \qquad E_n \propto \frac{1}{n^2}$$

- Electron in a He atom

$$\psi \propto e^{-2r/a_0}$$

the energy doesn't agree with expt.

reason: missing interaction between the electrons

Lecture 2: Underlying principles of DFT

The variational principle
Many electrons
Hohenberg-Kohn theorem
Kohn-Sham approach

1998 Nobel Prize in Chemistry

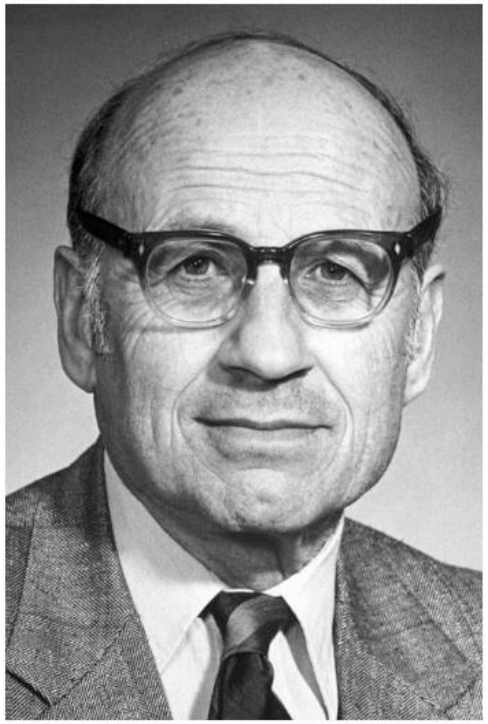

Photo from the Nobel Foundation archive.

Walter Kohn
Nobel Prize in Chemistry 1998

Born: 9 March 1923, Vienna, Austria

Died: 19 April 2016, Santa Barbara, CA, USA

Affiliation at the time of the award: University of California, Santa Barbara, CA, USA

Prize motivation: “for his development of the density-functional theory”

Prize share: 1/2

What is functional?

density functional theory

function $y = f(x)$ map x to y

functional map a function to a scalar

Example: expectation value of $\hat{H}$

$$F[\psi] = \int dV\, \psi^*(\vec{r})\, \hat{H}\, \psi(\vec{r})$$

ψ can be any function (doesn't have to be solution of S.E.)

IF ground state wavefunction $\hat{H}\psi_0 = \varepsilon_0 \psi_0$

$$F[\psi_0] = \int dV\, \psi_0^*(\vec{r})\, \underbrace{\hat{H}\, \psi_0(\vec{r})}_{\varepsilon_0 \psi_0(\vec{r})}$$

$$= \varepsilon_0 \underbrace{\int dV\, \psi_0^* \psi_0}_{1} = \varepsilon_0$$

Variational principle

For any normalised ψ

$F[\psi] \geq E_0$ ← ground state energy

Variational principle and S.E. are equivalent

Proof:

solving for ψ_0 = finding the minimum of $F[\psi]$

set the derivative of $F[\psi]$ to 0

$$0 = \frac{\delta}{\delta \psi^*(\vec{r})} \left[F[\psi] - \lambda \left\{ \int dV' \psi^*(\vec{r}') \psi(\vec{r}') - 1 \right\} \right]$$

rules: $\frac{\delta \psi^*(\vec{r}')}{\delta \psi^*(\vec{r})} = \delta(\vec{r} - \vec{r}')$ $\qquad \frac{\delta \psi(\vec{r})}{\delta \psi^*(\vec{r}')} = 0$

$$0 = \int dV' \, \delta(\vec{r} - \vec{r}') \, \hat{H} \, \psi(\vec{r}') - \lambda \int dV' \, \delta(\vec{r} - \vec{r}') \, \psi(\vec{r}')$$

$$\int dV' \, \delta(\vec{r} - \vec{r}') f(\vec{r}') = f(\vec{r})$$

$$= \hat{H} \psi(\vec{r}) - \lambda \psi(\vec{r})$$

$$\hat{H} \psi(\vec{r}) = \lambda \psi(\vec{r})$$

↑ interpret as energy

$\Rightarrow$ If we can find a wavefunction that minimizes $F[\psi]$, we don't need to solve S.E.

Systems with many electrons

$\psi(\vec{r})$

Need to consider el–el interaction

$\Psi(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N)$ for a N-el system

statistical interpretation: $|\Psi(\vec{r}_1, \vec{r}_2, \cdots \vec{r}_N)|^2 dV_1 dV_2 \cdots dV_N$ represents the probability of finding el. in volume dV_1 at position $\vec{r}_1$; el 2 in dV_2 at $\vec{r}_2$, ···

Many-el S.E.

$$\hat{H}\Psi(\vec{r}_1, \vec{r}_2, \cdots \vec{r}_N) = E\Psi(\vec{r}_1, \vec{r}_2, \cdots \vec{r}_N)$$

$\hat{H} =$

Many-electron Schrödinger's equation

$$\hat{H} = \sum_{i=1}^{N} -\frac{\hbar^2}{2m}\nabla_i^2 + \sum_{i=1}^{N} V(\vec{r}_i) + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{4\pi\varepsilon_0|\vec{r}_i-\vec{r}_j|} + \frac{1}{2}\sum_{I\neq J}\frac{e^2}{4\pi\varepsilon_0|\vec{R}_I-\vec{R}_J|}$$

kinetic energy

potential energy

e|- el interaction

nucleus- nucleus interaction

For He (2 electrons)

$$\left[-\frac{\hbar^2}{2m}\nabla_1^2 + -\frac{\hbar^2}{2m}\nabla_2^2 + V(\vec{r}_1) + V(\vec{r}_2) + \frac{e^2}{4\pi\varepsilon_0|\vec{r}_1-\vec{r}_2|}\right]\Psi(\vec{r}_1,\vec{r}_2)$$

$$= E\,\Psi(\vec{r}_1,\vec{r}_2)$$

Why is many-el S.E. impossible to solve?

1 el $\psi(\vec{r}_1)$ 3 coordinates

2 el $\Psi(\vec{r}_1, \vec{r}_2)$ 6 coordinates

⋮

N el 3N

M grids per coordinate

M^{3N} combinations

combinatorial growth

Applying the variational principle to many-electron SE

$$F[\bar{\Psi}] = \int dV_1 \cdots dV_N \, \bar{\Psi}^* \hat{H} \bar{\Psi} \geq E_0$$

E_0: ground state energy of the many-el system

In practice: don't want to know $\bar{\Psi}$

instead want to know E_0

knowing E_0 gives: forces, structures, phase diagram...

Can we get E_0 in a simpler way?

Hohenberg-Kohn theorem

$$F[\Psi] = F[\Psi[n]]$$

The ground state electron density uniquely determines the potential V(r) and thus all properties of the system including ψ and E_0

$$n(\vec{r}) = \int dV_2 \cdots dV_N \, |\Psi(\vec{r}, \vec{r}_2, \cdots, \vec{r}_N)|^2 + \int dV_1 dV_3 \cdots dV_N \, |\Psi(\vec{r}_1, \vec{r}, \vec{r}_3, \cdots \vec{r}_N)|^2 + \cdots + \int dV_1 dV_2 \cdots dV_{N-1} \, |\Psi(\vec{r}_1, \vec{r}_2, \cdots \vec{r}_{N-1}, \vec{r})|^2$$

Applying the variational principle to the electron density

$$E_0 \leq F[\Psi] = F[\Psi[n]] \equiv F[n]$$

F[n] is a functional of the electron density

$\Rightarrow$ density functional theory

E_0 can be obtained by minimizing F[n]

problem: don't know form of F[n]

Kohn-Sham approach

$F[n] \geq E_0$

How to write $F[n]$?

Four terms in many-el S.E.

① kinetic energy

② potential energy

③ el-el

~~④ nucleus-nucleus~~ E_{ion} doesn't depend on representation of electrons

I. rewrite ② & ③ as $n(\vec{r})$

② : $\int dV\, n(\vec{r})\, V(\vec{r})$

Part of ③ $\frac{e^2}{8\pi\varepsilon_0} \int dV \int dV' \frac{n(\vec{r})\, n(\vec{r}')}{|\vec{r} - \vec{r}'|}$ classical electrostatic Hartree energy

What's left : ① kinetic energy

and quantum mechanical effects

II. approximate kinetic energy of real interacting electrons by the kinetic energy of fictitious non-interacting electrons

Kohn - Sham electrons

KS orbitals $\{\psi_i(\vec{r})\}$

$$n = \sum_{i=1}^{N} |\psi_i(\vec{r})|^2$$

kinetic energy $\sum_{i=1}^{N} \int dV \, \psi_i^*(\vec{r}) \left[-\frac{\hbar^2}{2m} \nabla^2 \psi_i(\vec{r}) \right]$

III. Known parts F_{known} kinetic + potential + Hartree

Unknown $F_{unknown}$ XC exchange - correlation

$F_{xc}[\{\psi_i\}]$ True form not known

$$F[n] = \sum_{i=1}^{N} - \int dV\, \psi_i^*(\vec{r}) \frac{\hbar^2}{2m} \nabla^2 \psi_i(\vec{r})$$

$$+ \int dV\, n(\vec{r})\, V(\vec{r})$$

$$+ \frac{e^2}{8\pi\varepsilon_0} \int dV \int dV' \frac{n(\vec{r})\, n(\vec{r}')}{|\vec{r} - \vec{r}'|}$$

$$+ E_{xc}[n]$$

Applying the variational principle to KS electrons

$$0 = \frac{\delta}{\delta \psi^*(\vec{r}')} \left[F[\psi] - \sum_i \lambda_i \left\{ \int dV' |\psi_i(\vec{r})|^2 - 1 \right\} \right]$$

$$F[n] = \textcircled{1} \sum_{i=1}^{N} - \int dV \, \psi_i^*(\vec{r}) \frac{\hbar^2 \nabla^2}{2m} \psi_i(\vec{r}) + \textcircled{2} \int dV \, n(\vec{r}) V(\vec{r}) + \textcircled{3} \frac{e^2}{8\pi\varepsilon_0} \int dV \int dV' \frac{n(\vec{r}) n(\vec{r}')}{|\vec{r} - \vec{r}'|} + \textcircled{4} E_{xc}[n]$$

Term ①

$$\frac{\delta}{\delta \psi_i^*(\vec{r}')} \sum_j \int dV' \, \psi_j^*(\vec{r}') \left[-\frac{\hbar^2}{2m} \nabla^2 \psi_j(\vec{r}') \right]$$

$$= \sum_j \int dV' \, \delta_{ij} \delta(\vec{r} - \vec{r}') \left[-\frac{\hbar^2}{2m} \nabla^2 \psi_j(\vec{r}') \right]$$

$$= -\frac{\hbar^2}{2m} \nabla^2 \psi_i(\vec{r})$$

$$\frac{\delta \psi_j^*(\vec{r}')}{\delta \psi_i^*(\vec{r})} = \delta_{ij} \delta(\vec{r} - \vec{r}')$$

$$\frac{\delta \psi_j(\vec{r}')}{\delta \psi_i^*(\vec{r})} = 0$$

Term ②

$$\frac{\delta}{\delta \psi_i^*(\vec{r})} \int dV' \sum_j \psi_j^*(\vec{r}') \psi_j(\vec{r}') V(\vec{r}')$$

$$= \int dV' \sum_j \delta_{ij} \delta(\vec{r} - \vec{r}') \psi_j(\vec{r}') V(\vec{r}') = \psi_i(\vec{r}) V(\vec{r})$$

Term ③

$$\frac{\delta}{\delta \psi_i^*(\vec{r})} \frac{e^2}{4\pi\varepsilon_0} \int dV' \int dV'' \frac{\left(\sum_j \psi_j^*(\vec{r}') \psi_j(\vec{r}')\right)\left(\sum_k \psi_k^*(\vec{r}'') \psi_k(\vec{r}'')\right)}{|\vec{r}' - \vec{r}''|}$$

$$= \underbrace{\frac{e^2}{8\pi\varepsilon_0} \int dV' \frac{n(\vec{r}')}{|\vec{r} - \vec{r}'|}}_{V_H} \psi_i(\vec{r})$$

Term ④

$$\frac{\delta}{\delta \psi_i^*(\vec{r})} E_{xc}[n] = \int dV' \frac{\delta E_{xc}}{\delta n(\vec{r}')} \frac{\delta n(\vec{r}')}{\delta \psi_i^*(\vec{r})}$$

$$= \underbrace{\left| \frac{\delta E_{xc}}{\delta n} \right|}_{V_{xc}(\vec{r})} \psi_i(\vec{r})$$

Kohn – Sham Equation

$$-\frac{\hbar^2}{2m} \nabla^2 \psi_i(\vec{r}) + V(\vec{r}) \psi_i(\vec{r}) + v_H(\vec{r}) \psi_i(\vec{r}) + v_{xc}(\vec{r}) \psi_i(\vec{r})$$

$$= \varepsilon_i \psi_i(\vec{r})$$

Summary

$$\hat{H}\Psi(\vec{r}_1, \vec{r}_2, \cdots \vec{r}_N) = E\Psi(\vec{r}_1, \vec{r}_2, \cdots \vec{r}_N)$$

Many-body Schrödinger's equation

$$\hat{H} = \sum_{i=1}^{N} -\frac{\hbar^2}{2m}\nabla_i^2 + \sum_{i=1}^{N} V(\vec{r}_i) + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{4\pi\varepsilon_0|\vec{r}_i - \vec{r}_j|} + \frac{1}{2}\sum_{I\neq J}\frac{e^2}{4\pi\varepsilon_0|\vec{R}_I - \vec{R}_J|}$$

Variational principle: turn solving the equation into minimizing energy

$$F[\psi] \geq E_0$$

Hohenberg-Kohn theorem: use the electron density as the variable instead of the many-body wavefunction

$$F[n] \geq E_0 \quad \text{but don't know } F[n]$$

Kohn-Sham equation: use non-interacting electrons instead of the real interacting system

Lab 2: Quantum Espresso Input and Output for Molecules

https://imperialcollegelondon.github.io/MSE404-MM/

When you log in, you should have a data folder in your home directory, please store your files in this folder.